



Unit 4 · Outcome 1 · Climate Change

The slow snowfall that locks carbon away

KK03

VCE Environmental Science · Units 3 & 4 · Exam study summary

Off the coast at Cape Wirreanda Climate Observatory, billions of microscopic plants drift in the sunlit surface of the Southern Ocean. They breathe in carbon dioxide all day. When they die, they sink — a constant, silent fall of carbon drifting to the seafloor. Some of that carbon will be back in the air within a year. Some of it won't see daylight again for a million.

Digital Vector EnviroSci · single-topic mastery summary — clean explanations, comparison tables, visual aids, exam traps, command words and worked answers.



THE BIG IDEA

02 Sequestration = capture, then store

Carbon sequestration is the process of capturing carbon dioxide from the atmosphere and storing it in a reservoir — a place that holds carbon out of the air. Plants do it. Oceans do it. Soil does it. The whole point is that carbon leaves the atmosphere and goes somewhere it can stay.

Every reservoir that holds carbon is either a sink or a source, depending on which way the carbon is flowing on balance:

Absorbs more carbon than it releases. Net flow is out of the atmosphere. A young growing forest, a healthy ocean, a peat bog — all sinks.

Releases more carbon than it absorbs. Net flow is into the atmosphere. A burning forest, a drained wetland, a coal-fired power station — all sources.

✓ TIP

Watch the word "net" The same forest breathes carbon in (photosynthesis) and out (respiration, decomposition) every single day. Calling it a sink or a source is a statement about the balance over time, not about whether it ever releases carbon. Sequestration is only happening when the net flow is into storage.

◆ ANALOGY

🪣 Analogy · the leaky bucket Picture the atmosphere as a bucket of water. Sources are taps pouring in; sinks are drains letting water out. Sequestration is a drain. The bucket level (atmospheric CO₂) only stays steady when the drains keep up with the taps. For most of Earth's history they did — until humans opened a tap the natural drains were never built to match.

THE CENTRAL IDEA

03 Carbon moves at two very different speeds

This is the heart of the dot point. Carbon is sequestered through processes that operate on wildly different timescales. VCAA splits them into two:

Short-term (< 100 years)		Long-term (> 1000 years)
Also called	The fast / biological carbon cycle	The slow / geological carbon cycle
How carbon is stored	Living and recently-dead matter — leaves, wood, animals, soil humus, CO ₂ dissolved in surface seawater	Buried, compressed, mineralised — fossil fuels, limestone & other carbonate rock, deep-ocean sediment, deep peat
Key processes	Photosynthesis ↔ respiration, decomposition, surface ocean–air exchange	Burial & fossilisation, sedimentation, shell/carbonate formation, the biological pump to the deep ocean
Carbon comes back...	within a season to a human lifetime	over thousands to millions of years
Reservoir size / speed	Smaller stores, carbon turns over quickly	Enormous stores, carbon barely moves

◆ KEY INSIGHT

The single sentence that earns the mark Short-term sequestration cycles carbon between the atmosphere, living things and the surface ocean over up to ~100 years; long-term sequestration locks carbon into rock, fossil fuels and the deep ocean for more than 1000 years. The difference is how long the carbon is removed from the atmosphere.

◆ ANALOGY

Analogy · two bank accounts Short-term carbon is your everyday account — money flows in and out constantly, available within days. Long-term carbon is a term deposit locked for a thousand years — once it's in, it's gone from circulation. Burning coal is like smashing open a millennia-old term deposit and spending the lot in an afternoon. The everyday account simply can't absorb that much cash at once.

ON LAND

04 How land sequesters carbon

On the ranges behind Cape Wirreanda, carbon is captured and stored in three main ways — two fast, one slow:

Short-term. Photosynthesis pulls CO₂ from the air and builds it into leaves, stems and trunks. Stored for the life of the plant — years to centuries.

Short-term. Dead leaves and roots are part-decomposed into humus. Carbon-rich soil can hold it for decades, sometimes longer if undisturbed.

Long-term. In waterlogged bogs, dead plants can't fully rot. Carbon piles up for millennia — and over millions of years, buried and compressed, becomes coal.

◆ ANALOGY

Analogy · the compost bin that never finishes A peat bog is a compost bin permanently underwater. Decomposers need oxygen to finish the job, and waterlogged peat has almost none — so the dead plants get stuck halfway, half-rotted, piling up for thousands of years. That "unfinished compost" is one of the largest carbon stores on land. Drain the bog and you let the oxygen back in: decomposition restarts and the sink flips into a source.

⚠ WATCH OUT

Common slip Students often write that "trees store carbon forever." They don't — biomass carbon is short-term. When the tree dies and decomposes or burns, most of that carbon returns to the air within decades. Only the small fraction that becomes deep peat or eventually fossilises is long-term.

IN WATER

05 How the ocean sequesters carbon

The ocean is Earth's biggest active carbon sink. At Cape Wirreanda, the cold Southern Ocean swallows carbon through three mechanisms — the first fast, the next two reaching into the slow cycle:

Short-term. CO₂ simply dissolves into surface water — and cold water dissolves more, which is why the Southern Ocean is such a strong sink. Released again when water warms.

Long-term. Phytoplankton photosynthesise at the surface. When they die they sink as "marine snow" to the deep ocean, carrying carbon away from the air for centuries.

Long-term. Shellfish, corals and plankton build shells of calcium carbonate. These sink, accumulate as sediment, and over ages become limestone — carbon locked in rock.

✓ **TIP**

Blue carbon — worth a name-drop Coastal plants — mangroves, seagrass and saltmarsh — sequester carbon faster per hectare than rainforest, and bury much of it in waterlogged sediment where it stays for thousands of years. This is called blue carbon, and protecting it is a favourite exam example of a high-value, low-cost mitigation strategy.

⚠ **WATCH OUT**

The acidification trade-off When CO_2 dissolves in seawater it forms carbonic acid, lowering the ocean's pH — ocean acidification. So the same sink that helps the climate harms the shellfish and corals that build the carbonate store. A strong answer can note that the ocean sink is not "free."

INTERACTIVE

06 Fast loop, slow loop

Here are both cycles side by side. Carbon particles flow through a fast loop (air → plants/ocean surface → back to air) and a slow loop (down into burial and deep storage). Drag the timescale slider and watch what happens: as you speed up time, the fast loop blurs into a constant churn while the slow loop barely creeps. Press the buttons to send a pulse of carbon down either path.

Notice how little carbon makes it into the slow loop compared with the churning fast loop. That trickle into long-term storage is what kept atmospheric CO_2 roughly stable for thousands of years — and it's exactly the trickle that fossil-fuel burning overwhelms.

THE CLIMATE LINK

07 Why this dot point is really about fossil fuels

Long-term sequestration took carbon out of the air extremely slowly over hundreds of millions of years — dead plants and plankton buried, compressed and turned into coal, oil and gas. That's the planet's long-term carbon vault.

Burning fossil fuels does the reverse, but at a geological instant. Carbon that took 300 million years to lock away is released in 300 years of industry. The natural sinks — growing forests, the ocean solubility and biological pumps — can only sequester a fraction of it back fast enough.

◆ **KEY INSIGHT**

The imbalance in one line The rate at which humans release long-term carbon (combustion) hugely exceeds the rate at which natural sequestration can return it to storage. The surplus builds up in the atmosphere — and that surplus is the enhanced greenhouse effect you'll meet in KK04.

◆ ANALOGY

Analogy · the bath that overflows The natural drains (sinks) were sized for a slow trickle of carbon over millions of years. Fossil-fuel burning opens a fire-hose into the same bath. The drains haven't shrunk — they just can't cope with the new inflow, so the level rises. Climate change is the bath overflowing, not the plug being pulled.

✓ TIP

Connect it forward This is why protecting and restoring sinks (reforestation, blue carbon, peat) and engineered carbon capture & storage are mitigation options in KK11 — they all try to push carbon back down the slow loop faster.

DON'T LOSE EASY MARKS

08 Exam traps to dodge

⚠ WATCH OUT

Trap 1 · "sequestration = emission" Sequestration is carbon going into storage (a sink). Emission/combustion is carbon coming out. They are opposites — don't use them interchangeably.

⚠ WATCH OUT

Trap 2 · forgetting the timescale numbers The dot point names the figures: short-term < 100 years, long-term > 1000 years. Quoting them shows you know the exact distinction. Vague "fast and slow" answers cap your marks.

⚠ WATCH OUT

Trap 3 · land OR water, not both If a question says "in land and water," you need an example from each. A great forest answer alone won't get full marks on a both-realms question.

⚠ WATCH OUT

Trap 4 · treating a sink as permanent Sinks can flip to sources: a burnt forest, a drained peat bog, a warming ocean. If asked to evaluate a sink, mention what could reverse it.

⚠ WATCH OUT

Trap 5 · "the ocean just dissolves CO₂" That's only the solubility pump (short-term). The deep, long-term storage comes from the biological pump and carbonate burial. Name the right mechanism for the timescale the question asks about.

DECODE THE QUESTION

09 Command words

◆ COMMAND WORDS

The command word tells you how much to write and what kind of thinking earns marks. Tap one: Each one demands a different response. Knowing the difference is worth marks before you've written a word of science.

Command word	What the marker wants
Identify / State	Name it — no explanation.
Describe	Give the features / what happens, in order.
Explain	Give the how / why — cause and effect.
Distinguish / Compare	State the difference (and similarity) explicitly.
Analyse	Break into parts and show how they relate.
Evaluate / Justify	Weigh strengths & limits, then a reasoned verdict.

Match your answer to the verb — the fastest marks in the exam.

WRITE LIKE A STATE-RANKER

11 P·E·L trainer



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

P·E·L WRITING

Strong extended answers follow Point · Explain · Link. Make a clear point, explain the mechanism, then link it back to the question (here: the carbon cycle / climate). Draft each part for the prompt below — your writing saves locally.

◆ KEY INSIGHT

Prompt Explain how a coastal mangrove forest sequesters carbon, and identify whether this is a short-term or long-term change in the carbon cycle. (3 marks)

P·E·L WRITING

P: A mangrove forest is a carbon sink that sequesters carbon in both its biomass and the sediment beneath it.
 E: Photosynthesis captures atmospheric CO₂ and fixes it into roots, trunks and leaves; when this material falls into the waterlogged, low-oxygen mud, decomposition is extremely slow, so the carbon accumulates and is buried rather than released.
 L: Because that buried "blue carbon" is removed from the atmosphere for more than a thousand years, it represents a long-term change in the carbon cycle.

SEE THE DIFFERENCE

12 Weak vs strong — same question

Distinguish between short-term and long-term carbon sequestration. In your answer, refer to one example of each. (4 marks)

Short-term sequestration is when carbon is stored for a short time and long-term is when it is stored for a long time. Trees store carbon for short-term and the ocean stores it for long-term.

• **NOTE**

Why it loses marks: no timescales, "short/long time" just restates the words, and the examples are vague — the ocean does both, and trees aren't clearly tied to a mechanism. Likely 1/4.

Short-term sequestration stores carbon for less than ~100 years in the fast biological cycle — for example, photosynthesis fixing CO₂ into tree biomass, which returns to the air when the tree decomposes. Long-term sequestration stores carbon for more than 1000 years in the slow geological cycle — for example, marine organisms forming calcium-carbonate shells that sink and become limestone, locking carbon in rock.

• **NOTE**

Why it scores: both timescales quoted, both cycles named, and each example is tied to a specific mechanism in the correct realm. 4/4.

EXAM PRACTICE · SELF-MARK

13 Practice questions

★ **PRACTICE & SELF-MARK**

Five VCAA-style questions, — marks total. Try each one yourself first, then reveal the weak answer, strong answer and marking guide. Mark your own work honestly — your score builds in the dock.

- Sequestration = capturing CO₂ and storing it in a reservoir; a sink absorbs more than it releases, a source the reverse.
- Short-term (<100 yr): the fast/biological cycle — photosynthesis, respiration, decomposition, surface-ocean dissolution; stores in biomass, soil, surface water.
- Long-term (>1000 yr): the slow/geological cycle — burial & fossilisation, carbonate sediment/limestone, the biological pump to the deep ocean, deep peat.
- Land sinks: plant biomass & soil (short), peat & fossilisation (long).
- Water sinks: solubility pump (short), biological pump & carbonate burial & blue carbon (long).
- The climate link: combustion releases long-term carbon far faster than sequestration can recapture it — the surplus drives the enhanced greenhouse effect.

• **NOTE**

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KK03 UNIT 4 · CLIMATE CHANGE · DOT POINT 3

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That difference — between carbon that comes straight back and carbon that is locked away — is the whole of this dot point. It's called carbon sequestration, and understanding its two speeds is the key to understanding why burning fossil fuels is such a problem.

◆ KEY INSIGHT

The dot point, in plain words VCAA asks you to explain carbon sequestration in land and water that produces short-term (less than 100 years) and long-term (more than 1000 years) changes in the carbon cycle. So you need: what sequestration is, where it happens (land & water), and the two timescales it operates on.

EXAM TECHNIQUE

★ Worked answers — weak vs strong

The same question answered two ways. The strong column shows the moves that earn the marks.



The P·E·L answer shape — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Short-term sequestration is when carbon is stored for a short time and long-term is when it is stored for a long time. Trees store carbon for short-term and the ocean stores it for long-term. Why it loses marks: no timescales, "short/long time" just restates the words, and the examples are vague — the ocean does both, and trees aren't clearly tied to a mechanism. Likely 1/4.

✓ STRONG / MODEL ANSWER

Short-term sequestration stores carbon for less than ~100 years in the fast biological cycle — for example, photosynthesis fixing CO₂ into tree biomass, which returns to the air when the tree decomposes. Long-term sequestration stores carbon for more than 1000 years in the slow geological cycle — for example, marine organisms forming calcium-carbonate shells that sink and become limestone, locking carbon in rock. Why it scores: both timescales quoted, both cycles named, and each example is tied to a specific mechanism in the correct realm. 4/4.